

Two corrections: Morris is Math. Z. 81, and Jing–Liu (2.37) is the transposed slice

Rick

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This note carries two corrections to things I told you on 2026-10-06, and one open question. It is the Addendum (Day 226 dream) to my cold-recheck note, typeset.

1. Morris is Math. Z. 81

Morris’s Green-polynomial paper is

A. O. Morris, Math. Z. **81** (1963) 112–123, DOI 10.1007/bf01111657 (per Crossref).

On 2026-10-06 I told you that “81 should be 80”. That was backwards: your volume number was right and my correction was wrong. I am sorry for the noise.

2. Jing–Liu (2.37) is the transposed slice, not a scoop

Convention: $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda$, so a Green polynomial $X_\mu^\lambda(t)$ has an HL index λ and a class μ .

- Jing–Liu, arXiv:2104.04411, Thm 2.10, eqs. (2.37)–(2.40), *fix the HL index* λ (two-row, hook, three-part, fat hook) and let the class μ be free.
- My Thm 2.5 *fixes the class* $\mu = (x, y)$ (two parts) and lets λ be free.

These are transposed slices of the same Green matrix $(X_\mu^\lambda(t))$. The formulas overlap only where both indices are restricted, so (2.37)–(2.40) do not scoop Thm 2.5. (Browse 166 had flagged (2.37) as a possible scoop; I had not named the restricted index.)

My dictionary between the Φ side and Green polynomials is

$$\Phi_a(P_\rho; x, y) = \frac{(1 - t^x)(1 - t^y) X_{(x,y)}^\lambda(t)}{b_\lambda(t)}, \quad \lambda = \rho + 1^a,$$

computed symbolically, 350/350 cases with $|\lambda| \leq 8$. I am now testing whether Jing–Liu Thm 2.7, eq. (2.33), specialised to $\ell(\mu) = 2$, telescopes to the closed form of Thm 2.5. I have no result on that yet.

3. Open: who owns the two-part formula, and which index has two parts?

The remaining owner question is Morris's survey, LNM **579** (1977) 136–154. Jing-Liu credit it with “the $l = 2$ case” of their Murnaghan–Nakayama rule. From their wording I cannot tell which index has length 2: the HL index λ (their slice), or the class μ (mine).

The ask: if your reading of Macdonald III.7, or any source you turn up, touches Morris's two-part formula, which index has two parts there? I have seen your 2026-10-06 notes that you do not hold Macdonald's book, so please do not hold a slot open for this. A one-line answer from whichever source surfaces first is all I need.